Test Document: Project 4

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INTRODUCTION

This system, an evolution of a prior project, enhances data retrieval and display from large datasets. The original system processed geographical data from a CSV file, displaying a table of extreme zip codes for each state. The current system introduces several advancements, including a file converter that transforms the CSV file into a Zipcode Record (ZCR) file. The ZCR file, preserving original data integrity, adds additional information for user comprehension and dataset control. It features a fixed-size, semicolon-separated header providing high-level data information.

A new buffer system, designed for the ZCR file format, improves data usage efficiency and has been tested for compatibility and performance with the original program. A significant enhancement is the implementation of a primary key index based on the zipcode field value. Stored in a separate Index (IDX) file, this index can be loaded into memory for rapid data processing, reducing disk access needs. The IDX file enables swift specific record location by zipcode.

The system, operable from the command line, allows CSV or ZCR file name input for processing and offers CSV to ZCR file conversion functionality. It can locate a record based on user or command-line input (-Z flag). If the record is found, it is displayed; otherwise, a non-presence message is displayed. The system can access multiple records in a single execution.

This test document provides a comprehensive system usage guide. It includes detailed system feature and functionality descriptions, instructions for common tasks, and potential issue solutions. The document, organized into sections for easy navigation, is written in clear, concise language, with diagrams and examples for understanding aid. Regularly updated to reflect system changes and improvements, it is intended for end-users seeking effective system usage. Whether a new user or an experienced user seeking advanced feature information, this document is designed to maximize system usage.

TESTS

PROGRAM: proj3_original		
valid input values		
Zipcode .csv standard format rows organized	input: us_postal_codes.csv expected output: display formatted table with header showing state, easternmost, westernmost, northernmost, southernmost	State Easternmost Westernmost Northernmost Southernmost AA -82.2493 -82.2493 33.0364 33.0364 AK -176.6581 -130.0247 71.2346 51.8800 AL -88.3865 -84.9972 34.9804 30.2460 AP -155.7982 -85.8200 41.7206 19.1110
Zipcode .csv standard format rows random	input: us_postal_codes_ROWS_RANDOMIZED.csv expected output: display formatted table with header showing state, easternmost, westernmost, northernmost, southernmost	State Easternmost Westernmost Northernmost Southernmost AA -82.2493 -82.2493 33.0364 33.0364 AK -176.6581 -130.0247 71.2346 51.8800 AL -88.3865 -84.9972 34.9804 30.2460 AP -155.7982 -85.8200 41.7206 19.1110
invalid file types		
Zipcode .csv columns reorganized	input: us_postal_codes_place.csv expected output: display nothing	State Easternmost Westernmost Northernmost Southernmost
Zipcode .zcr Wrong file extension	input: us_postal_codes.zcr expected output: display nothing	State Easternmost Westernmost Northernmost Southernmost
incorrect user input		
Input other than a file	input: asdfdsf expected output: Failed to open file: asdfdsf	Failed to open file: asdfdsf
file does not exist		

file does not exist	input: this_is_a_test.csv expected output: Failed to open file: this_is_a_test.csv	Failed to open file: this_is_a_test.csv

PROGRAM: proj3_updated		
valid input values		
Zipcode .zcr standard format rows organized	input: us_postal_codes.zcr expected output: display formatted table with header showing state, easternmost, westernmost, northernmost, southernmost	State Easternmost Westernmost Northernmost Southernmost AA -82.2493 -82.2493 33.0364 33.0364 AK -176.6581 -130.0247 71.2346 51.8800 AL -88.3865 -84.9972 34.9804 30.2460 AP -155.7982 -85.8200 41.7206 19.1110
Zipcode .zcr standard format rows random	input: us_postal_codes_ROWS_RANDOMIZED.zcr expected output: display formatted table with header showing state, easternmost, westernmost, northernmost, southernmost	State Easternmost Westernmost Northernmost Southernmost AA -82.2493 -82.2493 33.0364 33.0364 AK -176.6581 -130.0247 71.2346 51.8800 AL -88.3865 -84.9972 34.9804 30.2460 AP -155.7982 -85.8200 41.7206 19.1110
Zipcode .csv standard format rows random	input: us_postal.csv expected output: display formatted table with header showing state, easternmost, westernmost, northernmost, southernmost	State Easternmost Westernmost Northernmost Southernmost AA -82.2493 -82.2493 33.0364 33.0364 AK -176.6581 -130.0247 71.2346 51.8800 AL -88.3865 -84.9972 34.9804 30.2460

		AP -155.7982 -85.8200 41.7206 19.1110
invalid file types		
Zipcode .zcr columns reorganized	input: us_postal_codes_place.zcr expected output: Invalid argument: stoi	Invalid argument: stoi State Easternmost Westernmost Northernmost Southernmost
incorrect user input		
Input other than a file	input: asdfdsfexpected output: File is not open. Cannot read from file.	Failed to open file: asdfdsf
file does not exist		
file does not exist	input: this_is_a_test.zcr expected output: File is not open. Cannot read from file.	File is not open. Cannot read from file.

PROGRAM: converter		
valid input values		
Zipcode .csv standard format rows organized	input: tes1.csv 1) expected output: Step 1: File < test1.csv > opened successfully Step 2: File < test1.zcr > opened successfully Step 3: Default header written to new .zcr Step 4: Records from .cvs written to new .zcr Step 5: Header updated for new .zcr File conversion complete…	Console: Step 1: File < test1.csv > opened successfully Step 2: File < test1.zcr > opened successfully Step 3: Default header written to new .zcr Step 4: Records from .cvs written to new .zcr Step 5: Header updated for new .zcr File conversion complete…
	2) Newly formatted .zcr with shared name but new extension	.zcr File:File Structure Type: ZCR;Version: 1.0;Header Record Size: 450;Record Size Integer Bytes: -1;Size Format Type: ASCII;Index File Name: primaryKey.idx;Index File Schema: Key, Offset;Record Count: 5;Fields Per Record: 6;Field 1: Zip Code, Int;Field 2: Place Name, String;Field 3: State, String;Field 4: County, String;Field 5: Lat, Double;Field 6: Long, Double;Primary Key: Zip Code; 42 501,Holtsville,NY,Suffolk,40.8154,-73.045142 544,Holtsville,NY,Suffolk,40.8154,-73.045139 1001,Agawam,MA,Hampden,42.0702,-72.622742 1002,Amherst,MA,Hampshire,42.3671,-72.4646

Zipcode .csv standard format rows random	input: tes2.csv 1) expected output: Step 1: File < test1.csv > opened successfully Step 2: File < test1.zcr > opened successfully Step 3: Default header written to new .zcr Step 4: Records from .cvs written to new .zcr Step 5: Header updated for new .zcr File conversion complete…	Console: Step 1: File < test1.csv > opened successfully Step 2: File < test1.zcr > opened successfully Step 3: Default header written to new .zcr Step 4: Records from .cvs written to new .zcr Step 5: Header updated for new .zcr File conversion complete…
	2) Newly formatted .zcr with shared name but new extension	.zcr fileFile Structure Type: ZCR;Version: 1.0;Header Record Size: 450;Record Size Integer Bytes: -1;Size Format Type: ASCII;Index File Name: primaryKey.idx;Index File Schema: Key, Offset;Record Count: 40933;Fields Per Record: 6;Field 1: Zip Code, Int;Field 2: Place Name, String;Field 3: State, String;Field 4: County, String;Field 5: Lat, Double;Field 6: Long, Double;Primary Key: Zip Code; 43 96737,Ocean View,HI,Hawaii,19.1002,-155.72645 97134,Oceanside,OR,Tillamook,45.4549,-123.96555 63701,Cape Girardeau,MO,Cape Girardeau,37.3169,-89.545934 88250,Hope,NM,Eddy,32.8156,-104.73

invalid file types		
Zipcode .csv columns reorganized	input: us_postal_codes_place.csv 1) expected output: Step 1: File < test1.csv > opened successfully Step 2: File < test1.zcr > opened successfully Step 3: Default header written to new .zcr Step 4: Records from .cvs written to new .zcr Step 5: Header updated for new .zcr File conversion complete…	Console: Step 1: File < test1.csv > opened successfully Step 2: File < test1.zcr > opened successfully Step 3: Default header written to new .zcr Step 4: Records from .cvs written to new .zcr Step 5: Header updated for new .zcr File conversion complete…
	2) .zcr header with no records	.zcr fileFile Structure Type: ZCR;Version: 1.0;Header Record Size: 450;Record Size Integer Bytes: -1;Size Format Type: ASCII;Index File Name: primaryKey.idx;Index File Schema: Key, Offset;Record Count: 40933;Fields Per Record: 6;Field 1: Zip Code, Int;Field 2: Place Name, String;Field 3: State, String;Field 4: County, String;Field 5: Lat, Double;Field 6: Long, Double;Primary Key: Zip Code;
Zipcode .zcr Wrong file extension	input: us_postal_codes.zcr expected output: Error: Input file is not a CSV file.	
Incorrect user input		
Input other than a file	input: asdfdsf expected output: Error message	Program began… terminate called after throwing an instance of 'std::out_of_range' what(): basic_string::substr: __pos (which is 18446744073709551615) > this->size() (which is 14) Aborted (core dumped)

PROGRAM: proj4_norm, proj4_norm_place, proj4_rand, proj4_rand_state		
valid input values		
User direct input	input: 501 expected output: the zipcode data	Zip Code: 501 Place Name: Holtsville State: NY County: Suffolk Latitude: 40.8154 Longitude: -73.0451
"-Z" commandline flag	input: -Z501 expected output: the zipcode data	Zip Code: 501 Place Name: Holtsville State: NY County: Suffolk Latitude: 40.8154 Longitude: -73.0451
User direct input: leading 0's	input: 00501 expected output: the zipcode data	Zip Code: 501 Place Name: Holtsville State: NY County: Suffolk Latitude: 40.8154 Longitude: -73.0451
"-Z" commandline flag leading 0's	input: -Z00501 expected output: the zipcode data	Zip Code: 501 Place Name: Holtsville State: NY County: Suffolk Latitude: 40.8154 Longitude: -73.0451

"-Z" commandline flagmultiple lines	input: ./proj4_rand -Z1585 -Z90221 -Z13628expected output: the zipcode data	Zip Code: 1585Place Name: West BrookfieldState: MACounty: WorcesterLatitude: 42.2441Longitude: -72.1511Zip Code: 90221Place Name: ComptonState: CACounty: Los AngelesLatitude: 33.8796Longitude: -118.217Zip Code: 13628Place Name: DeferietState: NYCounty: JeffersonLatitude: 44.0356Longitude: -75.6838
same input all variant subprograms		
proj4_norm	input: ./proj4_norm -Z99950 expected output: the zipcode data	Zip Code: 99950 Place Name: Ketchikan State: AK County: Ketchikan Gateway Latitude: 55.3422 Longitude: -131.648
proj4_norm_place	input: ./proj4_norm_place -Z99950 expected output: the zipcode data	Zip Code: 99950 Place Name: Ketchikan State: AK County: Ketchikan Gateway Latitude: 55.3422 Longitude: -131.648
proj4_rand	input: ./proj4_rand -Z99950 expected output: the zipcode data	Zip Code: 99950 Place Name: Ketchikan State: AK County: Ketchikan Gateway Latitude: 55.3422 Longitude: -131.648
proj4_rand_state	input: ./proj4_rand_state -Z99950expected output: the zipcode data	Zip Code: 99950Place Name: KetchikanState: AKCounty: Ketchikan GatewayLatitude: 55.3422Longitude: -131.648
incorrect user input		
User Input Incorrect Zipcode Entry	input: 54 expected output: No record found for zipcode: 54	No record found for zipcode: 54

"-Z" commandline flag Incorrect Zipcode Entry	input: -Z54 expected output: No record found for zipcode: 54	No record found for zipcode: 54
"-Z" commandline flag Multiple lines Invalid Zipcode Entry	input: ./proj4_rand_state -Z0 -Z2673 -Z97134 -Z66045 -Z99999 expected output: Valid zipcodes display and invalid show no record found	No record found for zipcode: 0 Zip Code: 2673 Place Name: West Yarmouth State: MA County: Barnstable Latitude: 41.6614 Longitude: -70.2363 Zip Code: 97134 Place Name: Oceanside State: OR County: Tillamook Latitude: 45.4549 Longitude: -123.965 Zip Code: 66045 Place Name: Lawrence State: KS County: Douglas Latitude: 38.959 Longitude: -95.2499 No record found for zipcode: 99999
incorrect user input		
User Input -Z entry	input: -Z544 expected output: No record found for zipcode: 0	No record found for zipcode: 0
"-Z" commandline flagNo flag on command line	input: 544expected output: No display	<Blank>
User Input Character other than digit	input: safdf expected output: No record found for zipcode: 0	No record found for zipcode: 0

CONCLUSION

The testing phase has provided valuable insights into the performance and reliability of the system. It has demonstrated robustness and efficiency in handling large datasets, with significant improvements in data retrieval and display compared to its predecessor. The introduction of the Zipcode Record (ZCR) file and the primary key index has notably enhanced the speed and precision of data processing.

However, like any complex system, there are areas where further testing and refinement are needed. While the system performed well under the conditions of the test cases, real-world usage may present scenarios that have not yet been anticipated. Therefore, it is crucial to continue refining the test cases to cover a broader range of possibilities.

While testing has confirmed the effectiveness of many features of the system, it has also highlighted areas for improvement. As the refinement and expansion of testing efforts continue, there is confidence that the reliability and performance of the system can be further enhanced.